

Anirudh Edupuganti

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SUMMARY

AI and backend engineer with 4+ years shipping production data and LLM systems across AWS, Azure, and GCP in healthcare, telecom, and pharmaceutical environments, including 0→1 agent and retrieval prototypes taken from POC to deployed service. Delivers production ETL and streaming pipelines on Spark, Snowflake, and Synapse, automates multi-cloud infrastructure with Terraform, and builds agentic systems end to end—RAG pipelines, MCP integrations, tool/function calling, multi-agent orchestration, and offline/online evaluation loops—in Python and TypeScript. Published researcher and patent holder, currently pursuing a PhD in Computer Science.

EXPERIENCE

AI / Data Platform Engineer (Data Engineer IV) | Elevance Health | Chicago, IL 02/2024 – 06/2025

- Owned a healthcare solutions platform end to end from problem framing through production, partnering directly with clinical and business stakeholders to define success metrics and ship incremental releases.
- Designed and deployed robust data integration workflows by building and optimizing SQL Server Integration Services (SSIS) packages.
- Authored and optimized a minimum of three complex stored procedures weekly using SQL Server Management Studio (SSMS).
- Created comprehensive project data maps while proactively anticipating potential challenges.
- Orchestrated weekly production releases across 3 environments via automated CI/CD, cutting release defects by 60% and reducing deployment time from 5hr to 2hrs.

Data Platform Engineer (Data Engineer IV) | Charter Communications | Stamford, CT 07/2022 – 01/2024

- Built cloud-native backend services and REST integrations orchestrating operational workflows across PEGA Systems and end-to-end Snowflake pipelines, serving 5 downstream consumers.
- Established robust AWS production environments and managed the complete lifecycle of cloud resources using Terraform.
- Built scalable data pipelines with Amazon EMR, Apache Spark, and AWS Glue.
- Tuned compute across EC2 and EMR clusters, cutting cloud spend by 30% and reducing job runtime by 20% through autoscaling, spot capacity, and partition strategy.
- Accelerated deployment timelines by 60% with automated CI/CD pipelines, containerized service deployments (Docker), and queue-based async processing for high-volume workloads.

Data Engineer / Developer | KVK Tech | Newtown, PA 12/2021 – 07/2022

- Led migration of a 5 TB data warehouse to Azure Synapse Analytics, cutting query latency by 80% and retiring 15 legacy pipelines with zero downtime.
- Engineered robust data integration solutions using Azure Data Factory.
- Executed comprehensive data processing pipelines on Google Dataproc integrating Spark and Hadoop.
- Developed custom Apache Spark functions and utilized DataFrames for complex feature engineering.
- Developed an FDA-compliant desktop application for comprehensive equipment tracking.

Data Engineer Intern | Indus Valley | Miamisburg, OH 09/2021 – 11/2021

- Designed and built automated ETL pipelines to ingest and process large-scale datasets.
- Classified and organized complex data assets, developing intuitive product dashboards using Tableau.
- Deployed and oversaw scalable Google Dataproc clusters and harnessed Google Cloud Functions for serverless applications.
- Provisioned and managed cloud infrastructure using Terraform.

Software Programming Intern | ISRO | Hyderabad, India 01/2019 – 04/2019

- Developed a Python-based deep learning solution for automated cloud cover detection.
- Engineered a robust data transformation pipeline for processing raw Sentinel satellite images.
- Successfully trained and evaluated the image recognition model achieving a 78% detection accuracy.

AI & AGENT ENGINEERING

Adaptive Multi-Index RAG System

- Architected a dual-routing Retrieval-Augmented Generation pipeline using semantic routing to direct queries between dense vector stores and structured SQL metadata, improving context retrieval accuracy by 40% as measured against a 200-question golden set with automated regression checks on every index change.
- Deployed the pipeline as a containerized service behind a REST API with queue-based ingestion in Python and Spark SQL, processing, chunking, and embedding thousands of unstructured documents daily for sub-second retrieval.

Secure Enterprise MCP Integration Server

- Developed a custom Model Context Protocol (MCP) server connecting enterprise data pipelines and infrastructure logs directly to Claude, enabling secure contextual data surfacing without exposing database credentials.
- Implemented resource schemas and tool-calling endpoints across the MCP lifecycle, allowing the model to autonomously analyze Terraform state files and SSIS execution logs to diagnose pipeline failures.

Autonomous Subagent Orchestration Workflow

- Built a hierarchical multi-agent system in which a supervisor agent dynamically delegates domain-specific tasks to specialized subagents equipped with custom API tools and skills.
- Used Claude Code to prototype and test agent state management, instrumenting each run with tracing and token/cost tracking to catch planning failures, and producing an automated workflow that fetches, processes, and summarizes multi-source data streams end to end.

EDUCATION

PhD Computer Science Andhra University, Vishakhapatnam, India	08/2023 – Present
Master's Degree in Computer Programming DePaul University, Chicago, IL	08/2020 – 05/2021
Bachelor's Degree in Computer Science Andhra University, Vishakhapatnam, India	08/2015 – 05/2019

SKILLS

- **Data Ecosystem:** Hadoop, Hive, Apache Spark, Spark SQL, DataFrames & RDDs, Kafka, Cassandra, NIFI, Sqoop (incremental imports), PEGA Systems; Avro, Parquet, ORC; Kryo/Java serialization
- **Cloud & Databases:** AWS (EMR, S3, EC2, Glue, SNS, Snowflake); Azure (Databricks, Synapse Analytics, Data Factory, ADLS, Cosmos DB); GCP (BigQuery, Dataproc, GKE, Cloud SQL, Cloud Functions); Terraform (multi-cloud IaC); MS SQL Server, MySQL
- **Languages & OS:** Python, TypeScript / Node.js, SQL, Scala, Bash / UNIX Shell Script; Linux, CentOS, Windows
- **Backend & Services:** REST / gRPC APIs, FastAPI, microservices, message queues (Kafka, SQS), Docker, Kubernetes (GKE), CI/CD (GitHub Actions), relational & document data modeling
- **Security & Observability:** Cloud IAM, KMS, Identity-Aware Proxy, Security Command Center, Google Cloud Operations suite, AWS SNS alerting, compliance policy implementation
- **Tools & Version Control:** Eclipse, Cursor, Kiro, Antigravity, Google AI Studio, ElevenLabs, R-Studio, Visual Studio, JIRA, GitHub, GIT
- **AI & Agentic Systems:** Retrieval-Augmented Generation (RAG) pipelines, Model Context Protocol (MCP) integrations, multi-agent & subagent architectures, tool/skill (function) calling, AI coding assistants (Claude Code, GitHub Copilot)
- **LLM Orchestration & Evaluation:** LangChain, LlamaIndex, AutoGen, prompt engineering & context-window optimization, LLM evaluation (LLM-as-a-judge, RAGAS, golden sets, offline/online evals, regression testing, human-in-the-loop review), fine-tuning (LoRA, QLoRA)
- **Vector & Context Engineering:** Pinecone, Chroma, Milvus, embedding models & semantic search, ETL pipelines for LLM ingestion (Spark SQL, SSIS)
- **LLMOps & Deployment:** model serving & API gateways, Infrastructure as Code (Terraform), AI system observability (LangSmith, Langfuse, OpenTelemetry tracing, cost/token tracking, latency & quality dashboards)

PUBLICATIONS & PATENTS

- **Papers:** Adaptive IoT AI Pest Control System with LLM Integration (Presented at ICICT 2026, London, UK; Paper ID: 842); Artificial Intelligence, IoT, and Digital Literacy for Sustainable Farming and Hunger Mitigation; Digital Roadmap for attaining sustainable development goals.
- **Patents:** Real-time biomedical image analysis (Approved); 3D biomedical image reconstruction (Approved).

CERTIFICATIONS

- IBM AI Engineering Professional Certificate (Coursera); Applied Generative AI Specialization (Simplilearn); Microsoft Certified: Fabric Analytics Engineer Associate (DP-600); Anthropic Claude 101, Claude Code 101, Claude Platform 101; Pega System Architecture & NBA Models (Pegasystems).
- Additional Coursework: Machine Learning & Deep Learning Foundations (Great Learning), CS50x (Harvard University), Data Engineering Hands-On (Udemy), Design Database with PostgreSQL (Codecademy), Research Methodologies (Queen Mary University), Python Basics (Simplilearn), Cyber Security Basics.